



Learning Python - Sheet #13

Spacewar #4

We need to add enemies in our game, so that our spaceship has something to shoot at.



Let's download the picture of an asteroid. Name it **meteorite.png**. For example, you can use this one: https://www.flaticon.com/free-icon/meteorite_3369682

Then update your file (**spacewar.py**) to add the following.

```
asteroid = pygame.image.load('../resources/meteorite.png')
asteroid = pygame.transform.scale(asteroid, (128, 128))
asteroids = []
new_asteroid_in = 0
NEW_ASTEROID_EVERY = 50
```

In this sheet, we'll use a new concept: a class. This allows to group together some data (here the asteroid position, speed and life) with the code which is using it (move and draw).

```
class Asteroid:

    def __init__(self):
        self.x = screen.get_width()
        self.y = random.randint(-asteroid.get_height(), screen.get_height())
        self.vx = random.randint(1, 10)
        self.vy = random.randint(-self.vx, self.vx)
        self.life = random.randint(1, 10)

    # Returns False if the asteroid disappeared from the screen
    def move(self):
        self.x -= self.vx
        self.y += self.vy
        if self.x + asteroid.get_width() < 0:
            return False
        if self.y + asteroid.get_height() < 0 or self.y > screen.get_height():
            return False
        return True

    def draw(self, screen):
        screen.blit(asteroid, (self.x, self.y))
```

 **Pay attention:** Look at the values chosen in **init**. Do you understand what they do and why they were chosen this way? What do you think would happen for different choices?

We still need functions outside of the class to create the asteroids and loop on them:

```
def new_asteroid(asteroids):
    global new_asteroid_in
    if new_asteroid_in == 0:
        asteroids.append(Asteroid())
        new_asteroid_in = NEW_ASTEROID_EVERY
    else:
        new_asteroid_in -= 1

def move_asteroids(asteroids):
    asteroids = [a for a in asteroids if a.move()]
```

Finally, change the function draw and your game loop (add the calls to the new functions and draw the missiles):

```
screen.blit(spaceship, (x, y))
for a in asteroids:
    screen.blit(asteroid, (a.x, a.y))
for m in missiles:
    (...)
    move_stars(stars)
    new_asteroid(asteroids)
    move_asteroids(asteroids)
    draw()
```

